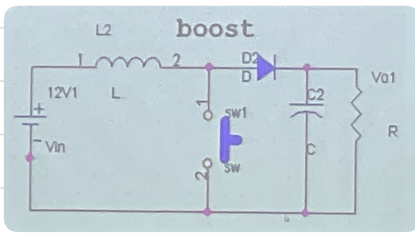


课堂笔记整理 4.21

DC-DC 升压电路



① 开关导通时, SW 闭合,

这时电感两端电压约为 $V_L = V_{in}$

电感电流逐渐上升: $\frac{di_L}{dt} = \frac{V_{in}}{L}$

电感在储存磁场能量, 输出靠电容 C 给 R 供电,

这时输入电源给电感充电, 输出电容支撑负载

② 开关断开时,

由于电感电流不会突变, 即其右端电压升高,

二极管导通, 电感开始释放能量, 输入电源和

电感一起向输出端送能量,

这时输出端得到的是输入电压+电感释放

产生的附加电压, 即升压

练 8-4

$$i_L(0^-) = \frac{100}{10} = 10A, \quad i_L(0^+) = i_L(0^-) = 10A \quad \text{即 } i_L(0) = 10A,$$

$$V_L(0^-) = 0, \quad V_L(0^+) = V_L(0) = 0, \quad V_L(0) = 0$$

$$R = 5\Omega, \quad L = 1H, \quad C = \frac{1}{9}F,$$

$$\alpha = \frac{R}{2L} = 2.5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = 3, \quad \alpha < \omega_0$$

$$\text{为欠阻尼, } \omega_d = \sqrt{\omega_0^2 - \alpha^2} = 1.6583$$

则通解为

$$i_L(t) = e^{-2.5t} (A_1 \cos 1.6583t + A_2 \sin 1.6583t)$$

$$A_1 = 10$$

$$\text{又由 } \frac{di_L}{dt} = -2.5e^{-2.5t} (A_1 \cos 1.6583t + A_2 \sin 1.6583t) + 1.6583e^{-2.5t} (-A_1 \sin 1.6583t + A_2 \cos 1.6583t)$$

$$\text{令 } t=0, \text{ 则 } \frac{di_L}{dt}(0) = -2.5A_1 + 1.6583A_2$$

$$\text{又由 } L \frac{di_L}{dt} + Ri_L + V_L = 0, \quad \frac{di_L}{dt}(0) = -\frac{Ri_L(0) + V_L(0)}{L}$$

$$\text{代入得 } \frac{di_L}{dt}(0) = -50, \quad \text{则 } -50 = -2.5A_1 + 1.6583A_2$$

$$\text{解得 } A_2 = -15.076$$

$$\text{故 } i_L(t) = e^{-2.5t} (10 \cos 1.6583t - 15.076 \sin 1.6583t) A,$$

练 8-5

$$\alpha = \frac{1}{2RL} = 10, \quad \omega_0 = \frac{1}{\sqrt{LC}} = 10, \quad \alpha = \omega_0, \text{ 为临界阻尼.}$$

$$\text{则 } v(t) = (A_1 + A_2 t) e^{-10t}, \quad t > 0$$

$$\text{由 } v(0) = 0, \quad v(0) = (A_1 + A_2 \cdot 0) e^0 = A_1 = 0,$$

$$\text{故 } v(t) = A_2 t e^{-10t}$$

$$\frac{dv(0^+)}{dt} = \frac{i_L(0^+)}{C} = \frac{-50}{25} = -2V/s$$

$$A_2 = -2, \quad \text{则 } v(t) = (-2t) e^{-10t} V$$

练 8-6

$$\alpha = \frac{1}{2RL} = 0.25, \quad \omega_0 = \frac{1}{\sqrt{LC}} = 5, \quad \alpha > \omega_0$$

$$v(t) = A_1 e^{-2.5t} + A_2 e^{-10t}, \quad v(0) = 0, \quad \text{则 } A_1 + A_2 = 0, \quad \frac{dv(t)}{dt} = -1125,$$

$$\text{则 } \frac{dv}{dt} = -2.5A_1 e^{-2.5t} - 10A_2 e^{-10t},$$

$$\frac{dv}{dt}(0) = -2.5A_1 - 10A_2, \quad -1125 = -2.5A_1 - 10(-A_1)$$

$$\Rightarrow A_1 = -150, \quad A_2 = 150$$

$$\text{则 } v(t) = 150(e^{-10t} - e^{-2.5t}) \text{ V}, \quad t > 0$$

练 8-7

$$v(0) = \frac{1}{2H} \times 18 = 12 \text{ V}, \quad v(\infty) = 15 \text{ V}, \quad \alpha = \frac{R}{2L} = 2, \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2.5 \times \frac{1}{40}}} = 4$$

$$\alpha < \omega_0, \text{ 为欠阻尼, } \omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{16 - 4} = \sqrt{12} = 3.464$$

$$v(t) = v(\infty) + e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

$$v(t) = 15 + e^{-2t} (A_1 \cos 3.464t + A_2 \sin 3.464t)$$

$$\text{由 } 12 = 15 + A_1, \text{ 得 } A_1 = -3$$

$$\text{由 } \frac{dv}{dt}(0) = 0, \text{ 可求得 } A_2 = -1.732$$

$$\text{则 } v(t) = 15 - (3 \cos 3.464t + 1.732 \sin 3.464t) e^{-2t} \text{ V}, \quad t > 0$$

$$v_R(t) = RL \frac{dv}{dt} = \frac{1}{4} \frac{dv}{dt} = 3.464 e^{-2t} \sin 3.464t \text{ V}, \quad t > 0$$